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PROJECT REPORT ON

Entity Relationship Model of “TRAFFIC” *for the Partial Fulfilment of 3rd semester Course*

DATABASE MANAGEMENT SYSTEM

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ABSTRACT

Our Project is a desktop based application which is helpful to reduce the involved risk, it is highly desirable for bike-riders to use helmet. In this application we propose an approach for automatic detection of bike-riders without helmet using surveillance videos in real time. The proposed approach first detects bike riders from surveillance video using background subtraction. Then it determines whether bike-rider is using a helmet or not by using Machine Learning.

In order to evaluate our approach, we use HISTOGRAM OF ORIENTED GRADIENTS (HOG). Also, we present a consolidation report of violation reporting which helps improving reliability of the proposed approach. Automatic detection of bike-riders without helmet falls under broad category of in surveillance videos.

The proposed approach for real-time detection of bike-riders without helmet which work in two phases. In the first phase, we detect a bike-rider in the video frame. In the second phase, we locate the head of the bike-rider and detect whether the rider is using helmet or not.

In this Project, there are 2 main actors or users who are performing the tasks in between them. The first actor is ADMIN, who is responsible for managing rules and regulations and the person who is violating the rules or not wear helmet, admin produce challan (fine) to particular person. The second actor is DRIVER, the person who is riding bike or driving car. If driver or the person who is riding bike should be caught and he have to pay the fine.

This Project was developed to reduce the traffic and accident due to violation, easy to identify driver without helmet. Helping in traffic management.

If this idea can be deployed by the traffic and accident management, it will reduce accident and people will follow the rules and regulation. This will also save the time required to produce challan.

And these application is for safety of bike riders and to reduce accident.

1. INTRODUCTION TO ER MODEL

The Entity Relationship (ER) Model is a high-level conceptual data model that defines data elements and their relationships. It is widely used in database design to create a logical representation of the structure and relationships within a system.

The Entity-Relationship (ER) model was developed by Peter Chen in 1976 to provide a unified framework for data modeling. It became a foundational concept for relational database design, allowing for the representation of data structures and their relationships in a clear, abstract way, facilitating database design and other systems applications.

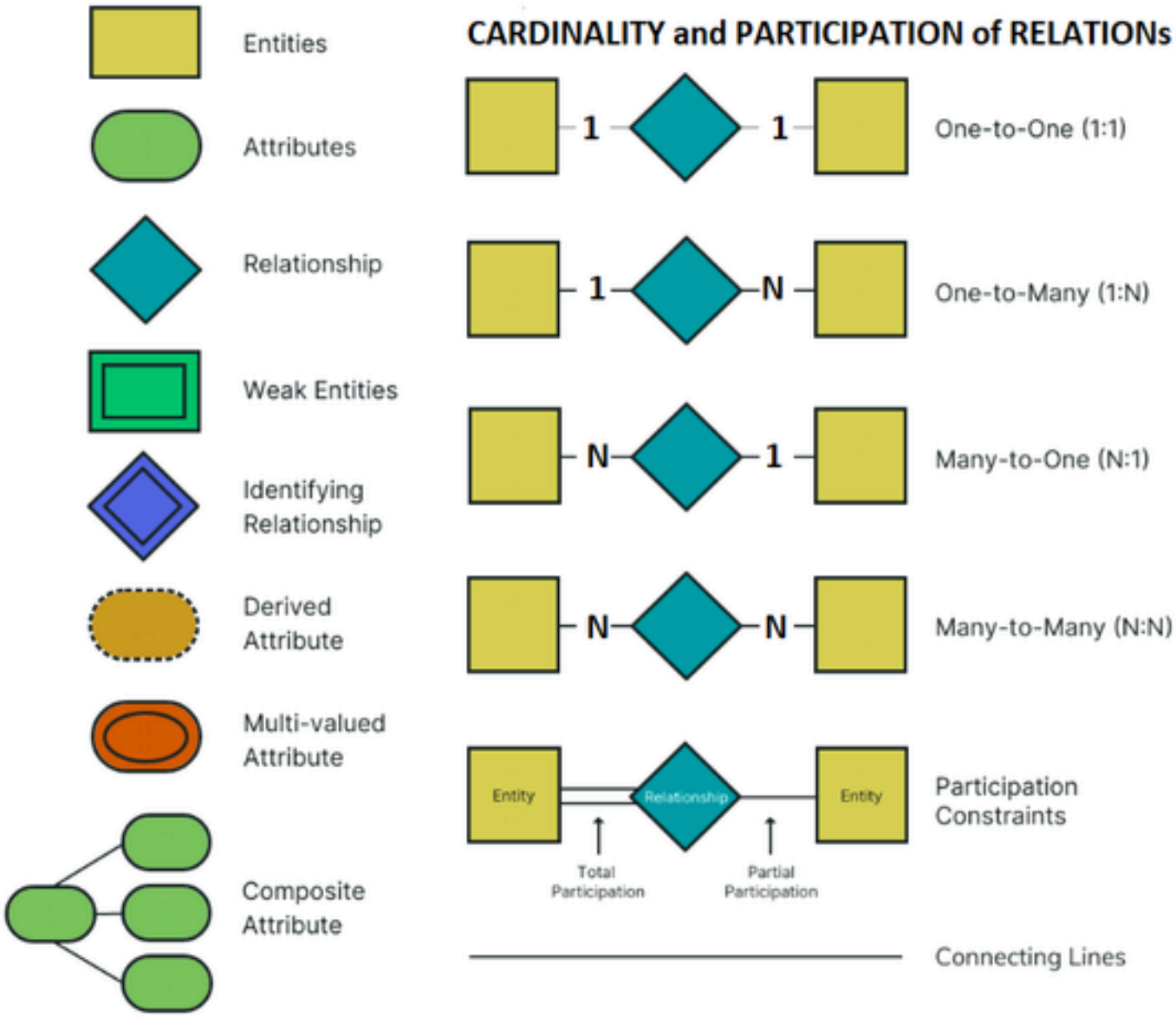
The main Components of ER Model are

- **Entities** represent real-world objects or concepts.
- **Attributes** describe the properties of entities.
- **Relationships** define how entities are connected to each other.

Characteristics of Model:

- ER model allows you to draw Database Design
- It is an easy to use graphical tool for modeling data Widely used in Database Design
- It is a GUI representation of the logical structure of a Database
- It helps you to identifies the entities which exist in a system and the relationships between those entities
- ER model is used to show the C*onceptual* schema of an organisation.
- Independent of specific data model or DBMS
- The model is later transformed into a *Logical* model (e.g. relational) on which the physical database is built

ERD Symbols and Notation



Entity-Relationship Diagrams (ERDs) are a fundamental part of database design and modeling. They represent the logical structure of databases. Here are the primary symbols and notations used in ERDs:

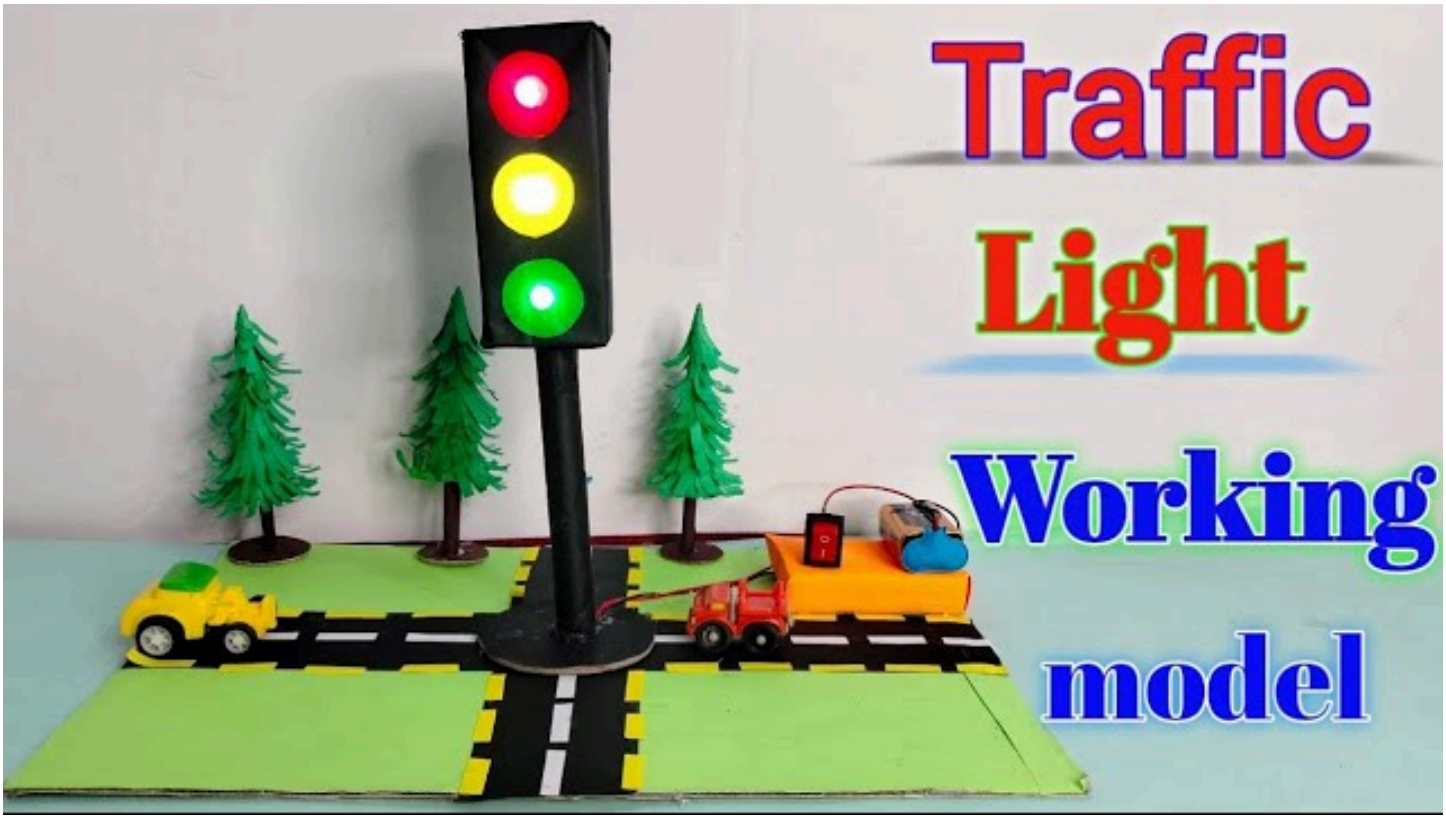
1. **Entities**: Represented by rectangles, entities are objects or concepts about which data is stored. Each entity has a name which is typically a noun, such as “Customer” or “Order”.
2. **Attributes**: Represented by ovals, attributes are the data we collect about the entities. An attribute may be a primary key (underlined), which uniquely identifies each instance of the entity.

3. **Relationships**: Depicted by diamonds, relationships show how two entities share information in the database. The name of the relationship is often a verb, such as "buys" or "owns".
4. **Connecting lines**: Solid lines are used to connect entities to attributes and entities to relationships, illustrating the linkage between them.
5. **Cardinality and modality**: These are the indicators of the nature of the relationship between entities. Cardinality refers to the number of instances of one entity that can or must be associated with each instance of another entity. Modality (or optionality) indicates whether or not an instance of an entity is mandatory and can be shown by either a circle (optional) or a bar (mandatory).
   - **One-to-One (1:1)**: A single instance of an entity is associated with a single instance of another entity.
   - **One-to-Many (1:N)**: A single instance of an entity is associated with many instances of another entity.
   - **Many-to-One (N:1)**: Many instances of an entity are associated with a single instance of another entity.
   - **Many-to-Many (N:N)**: Many instances of an entity are associated with many instances of another entity.
6. **Weak entities**: Sometimes represented by a double rectangle, a weak entity is an entity that cannot exist in the database without the existence of another entity (its owner).
7. **Identifying relationships**: Depicted by a double diamond, these represent the connection between a weak entity and its owner entity.
8. **Derived attribute**: Represented by a dashed oval, a derived attribute is one whose value is calculated from other attributes.
9. **Composite attribute**: If an attribute is made up of multiple components, it is shown as an oval with ovals inside it, indicating the composition.
10. **Multi-valued attribute**: Depicted by a double oval, a multi-valued attribute can have more than one value for a single entity (like multiple phone numbers for one person).
11. **Participation constraints**: Sometimes, there are constraints on how entities participate in relationships, which are depicted by placing either "partial" or "total" on the lines that connect entities to relationships

2. PROJECT INTRODUCTION

Road Traffic Safety refers to the methods and measures used to prevent road users from being killed or seriously injured. Typical roud users include pedestrians, cyclists, motorists, vehicle, passengers, horse riders and passengers of on-road public transport buses and trams. Road Safety

measures capacity more effectively ammed at improving the safety and flow of traffic utilizing traffic



1. Purpose

Games on Road Safety is mainly improvised for looking after the set off data of a region to manage the traffic along that area and implernent various useful technologies which are been required by various persones. It is mainly used for the purpose of games on road safety is to give the details which can be used and they can be implemented in their daily life. It is also mainly used for to create awareness among the people. The problems which have been occurred at their presence can be solved by these games on road safety

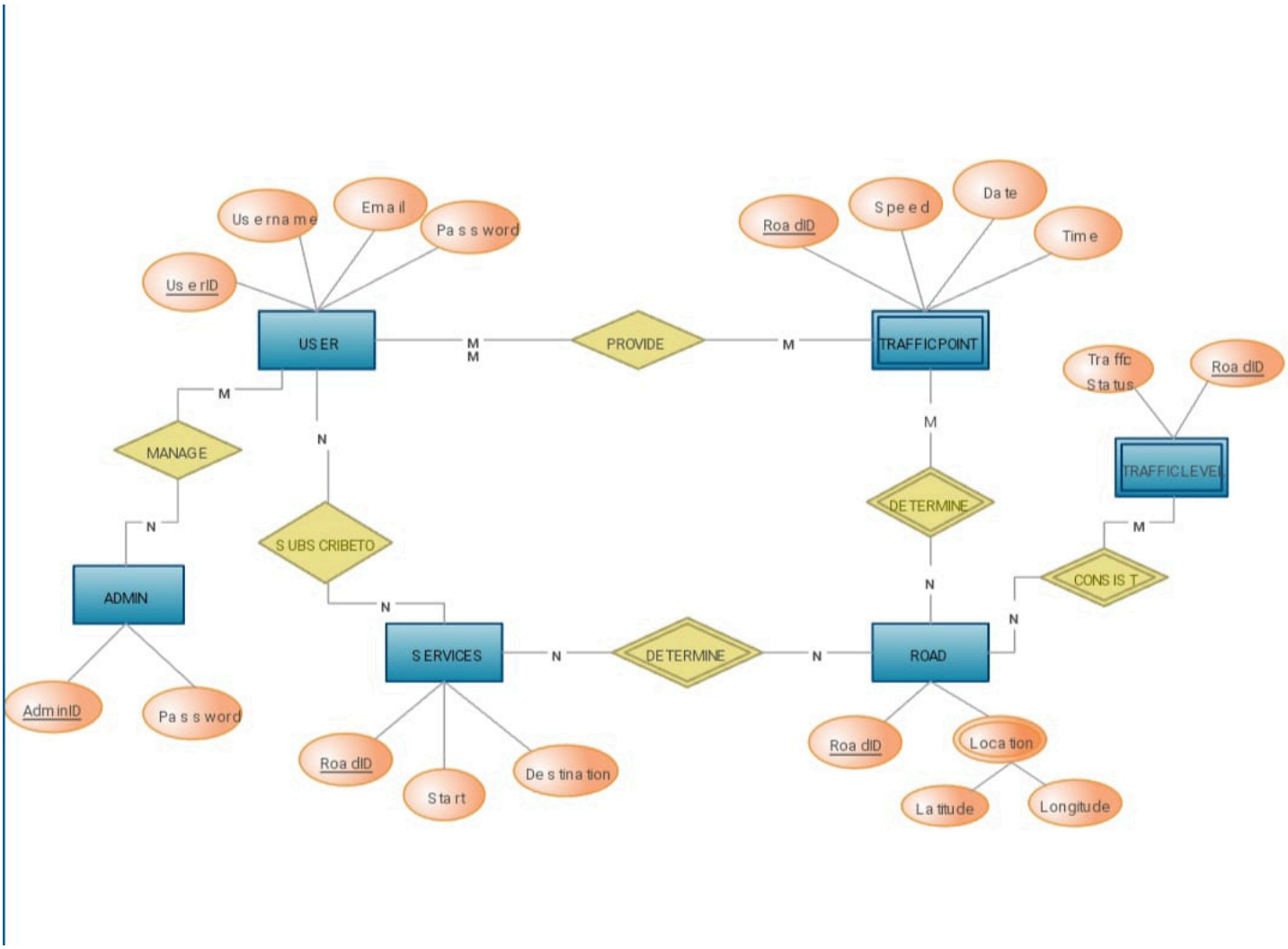
1. Scope

The Scope of this project is to create awareness among the people by conducting games like quizzes, identifying the images related to traffic rules, signals, directions in order to avoid the high rate of accidents. This project can be used for incident detection of a roadway.

3.ER DIAGRAM

The ER diagram represents the logical structure of the database. Below is an outline of the diagram's key components:

- **Entities**: traffic police,traffic,routes,length, Diversions, Vehicle type.



DESCRIPTION OF TRAFFIC MONITORING SYSTEM DATABASE:

- The details of Traffic is store into the Traffic tables respective with all tables

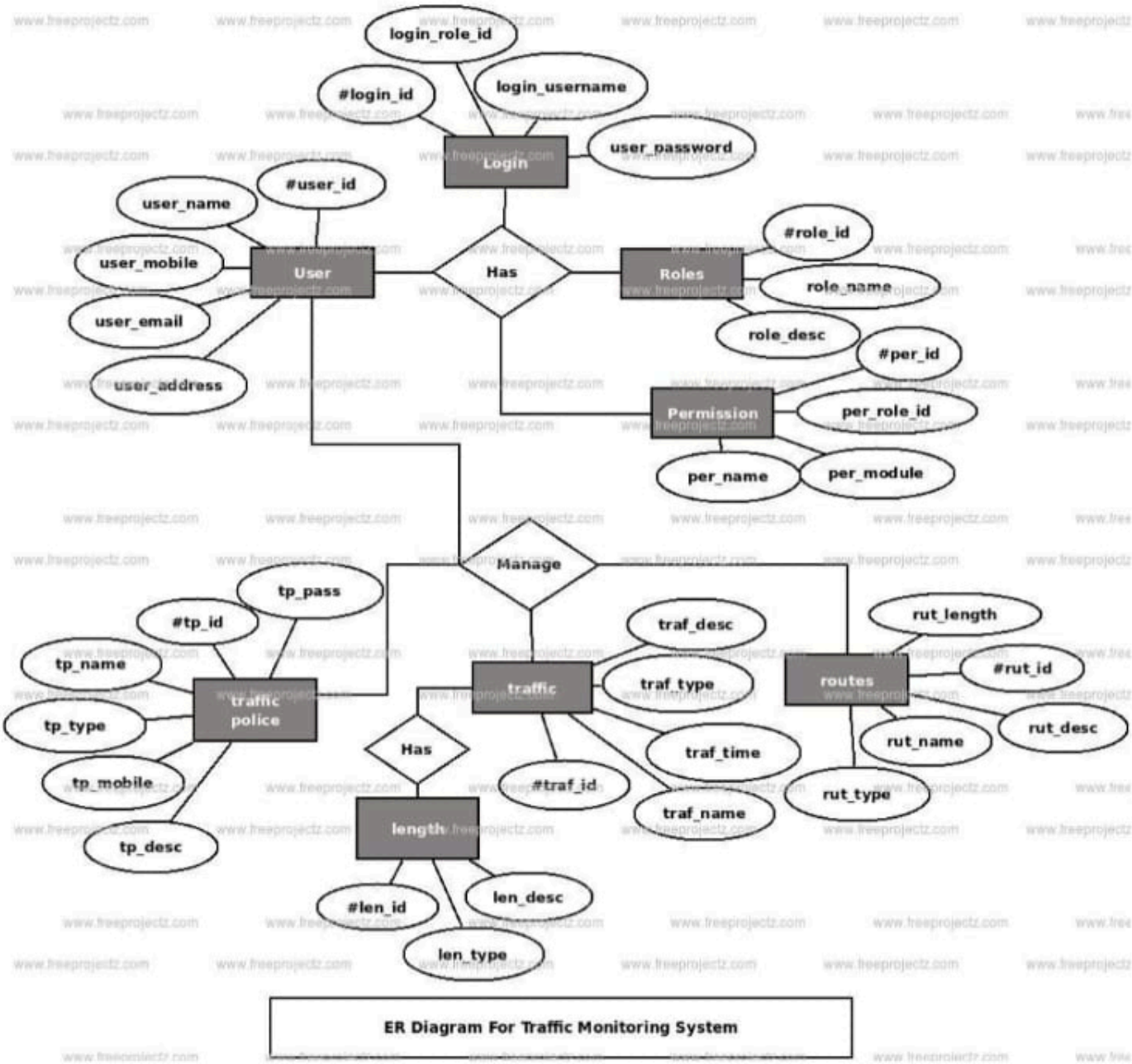
- Each entity (Vehicle Types, Length, Divertions, Routes, Traffic) contains primary key and unique keys.

- The entity Length, Divertions has binded with Traffic, Routes entities with foreign key

- There is one-to-one and one-to-many relationships available between Divertions, Traffic Polices, Vehicle Types, Traffic

- All the entities Traffic, Divertions, Length, Vehicle Types are normalized and reduce duplicacy of records

- We have implemented indexing on each tables of Traffic Monitoring System tables for fast query execution.



ER Diagram For Traffic Monitoring System

4. COMPONENTS DESCRIPTION

ENTITIES

***TRAFFC* MONITORING SYSTEM ENTITIES AND THEIR ATTRIBUTES:**

1. ***Traffic Entity*:**

Attributes of Traffic are traffic_id, traffic_name, traffic_type, traffic_description

2. ***Routes Entity*:**

Attributes of Routes are route_id, route_name, route_type, route_description

3. ***Length Entity***:

Attributes of Length are length_id, length_name, length_type, length_description

4. ***Traffic Polices Entity***:

Attributes of Traffic Polices are traffic_police_id, traffic_police_college_id, traffic_police_name, traffic_police_mobile, traffic_police_email, traffic_police_username, traffic_police_password, traffic_police_address

5. ***Divertions Entity***:

Attributes of Divertions are divertion_id, divertion_name, divertion_type, divertion_description

6. ***Vehicle Types Entity***:

Attributes of Vehicle Types are vehicle_type_id, vehicle_type_customer_id, vehicle_type_number, vehicle_type_description

ATTRIBUTES:

Attributes are described for each entity above. Primary keys uniquely identify each entity, and foreign keys establish relationships between entities.

RELATIONSHIP:

The ER (Entity-Relationship) model in the image depicts several entities related to traffic management and their relationships. Let's break down the key relationships:

1. Traffic & Routes:

- A relationship likely exists between Traffic and Routes.
- This could be a many-to-many relationship as multiple traffic instances can occur on a single route, and a single traffic instance can affect multiple routes.
- The relationship could be named "Affects" or "Occurs On."

2. Traffic & Diversions:

- Another relationship could exist between Traffic and Diversions.
- This could be a one-to-many relationship as a single traffic instance might lead to multiple diversions.
- The relationship could be named "Causes" or "Leads To."

3. Traffic & Vehicle Types:

- A relationship might exist between Traffic and Vehicle Types.

- This could be a many-to-many relationship as different types of vehicles can be involved in traffic, and traffic can affect different types of vehicles.
- The relationship could be named "Involves" or "Impacts."

4. Traffic & Traffic Polices:

- A relationship could exist between Traffic and Traffic Polices.
- This could be a many-to-many relationship as multiple traffic polices can manage a traffic instance, and a single traffic police can manage multiple traffic instances.
- The relationship could be named "Manages" or "Handles."

5. Routes & Length:

- A relationship might exist between Routes and Length.
- This could be a one-to-one relationship as each route would have a specific length.
- The relationship could be named "Has Length."

NOTE:

The exact nature of these relationships (e.g., cardinality, optionality) depends on the specific business rules and requirements.

The ER model can be further enhanced by adding attributes to the relationships, such as timestamps or additional details about the relationship.

5. CONCLUSION

Conclusions from research on traffic models using entity-relationship (ER) diagrams:

TRAFFIC DATA MODELING

Polynomial regression and multinomial logistic regression are two techniques used to model traffic data. The polynomial regression technique correctly categorizes data 91.3% of the time, while the multinomial logistic regression technique correctly categorizes data 90.09% of the time.

TRAFFIC MODEL SYNTAX

The syntax of traffic models can be meta-modeled in the ER diagrams formalism.

TRAFFIC FORECASTING

Neural network algorithms can be used to predict traffic in the coming days based on previous days' traffic data. The Feed-forward Levenberg-Marquardt method is the most accurate method, predicting traffic data with 81.59% accuracy.

TRAFFIC VARIABLES

Traffic lights and speed bumps are variables that cause different travel speeds.

ER diagrams are a graphical interface used in database management systems (DBMS) to create, modify, and optimize databases. They help simplify the process of defining relationships, establishing constraints, and creating tables.

6.BIBLIOGRAPHY

 Elmasri, R., & Navathe, S. B. (2016). *Fundamentals of Database Systems*. Pearson.

 Connolly, T., & Begg, C. (2015). *Database Systems: A Practical Approach to Design, Implementation, and Management*. Pearson.

 Online resources:

- ERDplus.com
- W3Schools Database Tutorials
- GeeksforGeeks ER Model